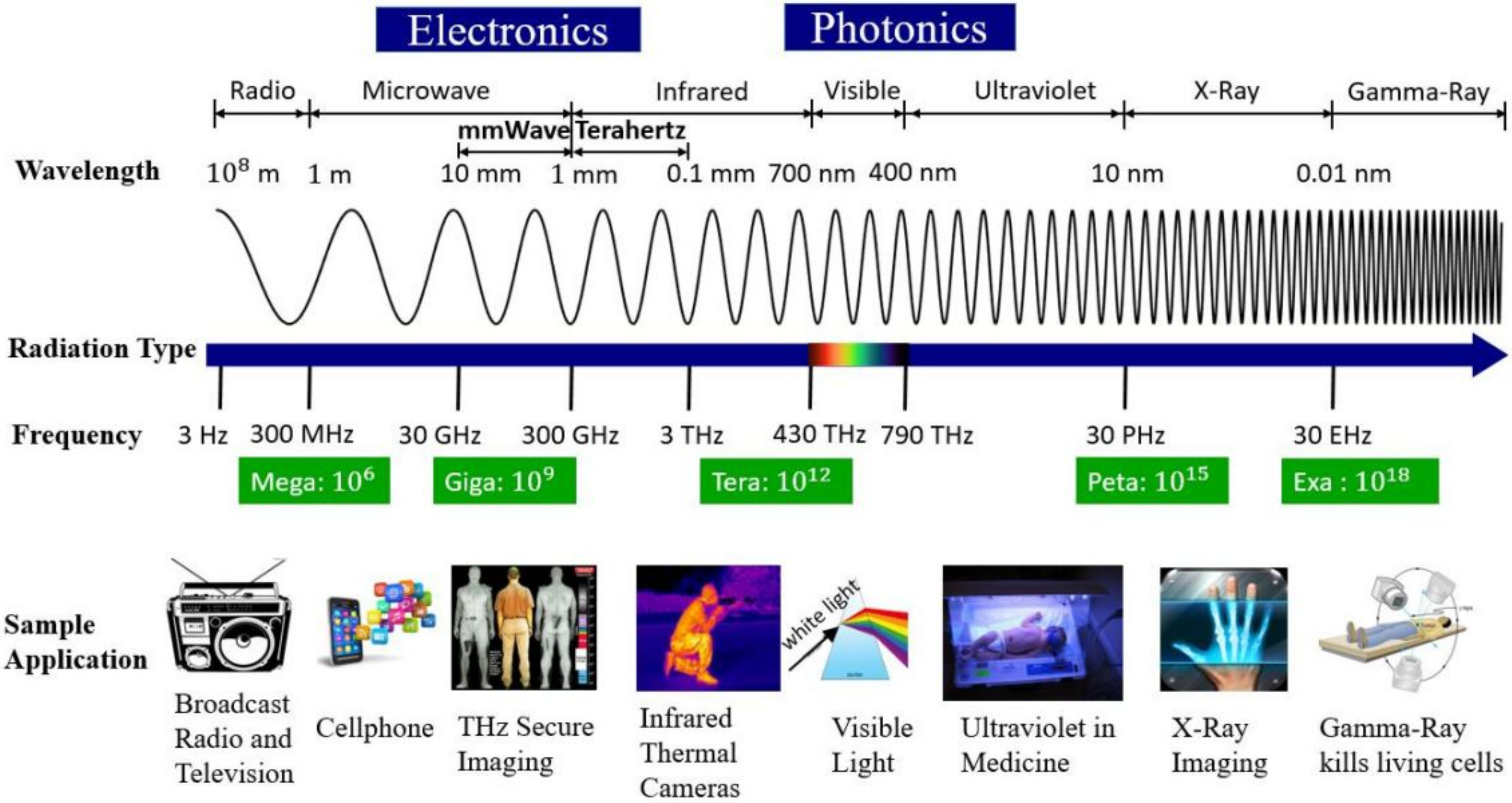
# mm Wave Communication IEEE 802.15.3C IEEE 802.15.3 MAC

# Electromagnetic Spectrum & Applications




[1] T. S. Rappaport, Y. Xing, O. Kanhere, S. Ju, A. Alkhateeb, G. C. Trichopoulos, A. Madanayake, S. Mandal, "Wireless Communications and Applications Above 100 GHz: Opportunities and Challenges for 6G and Beyond (Invited)," IEEE ACCESS, submitted Feb. 2019.

# Band designation

| Bands | Frequency range | Typical uses |
|---|---|---|
| L | 1 - 2 GHz | military telemetry, GPS, mobile phones (GSM), amateur radio. |
| S | 2 - 4 GHz | weather radar, surface ship radar, and some communications satellites, microwave ovens, radio astronomy, mobile phones, wireless LAN, Bluetooth, ZigBee, GPS, amateur radio. |
| C | 4 - 8 GHz | long-distance radio telecommunications. |
| X | 8 - 12 GHz | satellite communications, radar, terrestrial broadband, space communications, amateur radio. |
| Ku | 12 - 18 GHz | satellite communications. |
| K | 18 - 26.5 GHz | radar, satellite communications, astronomical observations. |
| Ka | 26.5 - 40 GHz | satellite communications. |
| Q | 33 - 50 GHz | satellite communications, terrestrial microwave communications, radio astronomy, automotive radar. |
| V | 50 - 75 GHz | millimetre wave radar research and other kinds of scientific research. |
| E | 60 - 90 GHz | UHF transmissions. |
| W | 75 - 110 GHz | satellite communications, millimeter-wave radar research, military radar targeting and tracking applications, and some non-military applications. |
| F | 90 - 140 GHz | SHF transmissions: Radio astronomy, microwave devices/communications, wireless LAN, most modern radars, communications satellites, satellite television broadcasting, DBS, amateur radio. |
| D | 110 - 170 GHz | EHF transmissions: Radio astronomy, high-frequency microwave radio relay, microwave remote sensing, amateur radio, directed-energy weapon, millimetre wave scanner. |

# Introduction

The rapid increase of mobile data and the use of smart phones are creating unprecedented challenges for wireless service providers to overcome a global bandwidth shortage.

- As today's cellular providers attempt to deliver high quality, low latency video and multimedia applications for wireless devices, they are limited to a carrier frequency spectrum ranging between 700 MHz and 2.6 GHz

# mm-wave

- Mm-Wave is a promising technology for future cellular systems. Since limited spectrum is available for commercial cellular systems, most research has focused on increasing spectral efficiency by using OFDM, MIMO, efficient channel coding
- Network densification has also been studied to increase area spectral efficiency, including the use of heterogeneous infrastructure (macro-, Pico-, femto cells, relays, distributed antennas) but increased spectral efficiency is not enough to guarantee high user data rates. The alternative is more spectrum.
- Millimeter wave (mm-Wave) cellular systems, operating in the 30-300GHz band, above which electromagnetic radiation is considered to be low (or far) infrared light, also referred to as terahertz radiation.

# Evolution of Wireless Technologies

1933

FM Radio
(~100 MHz)

1973

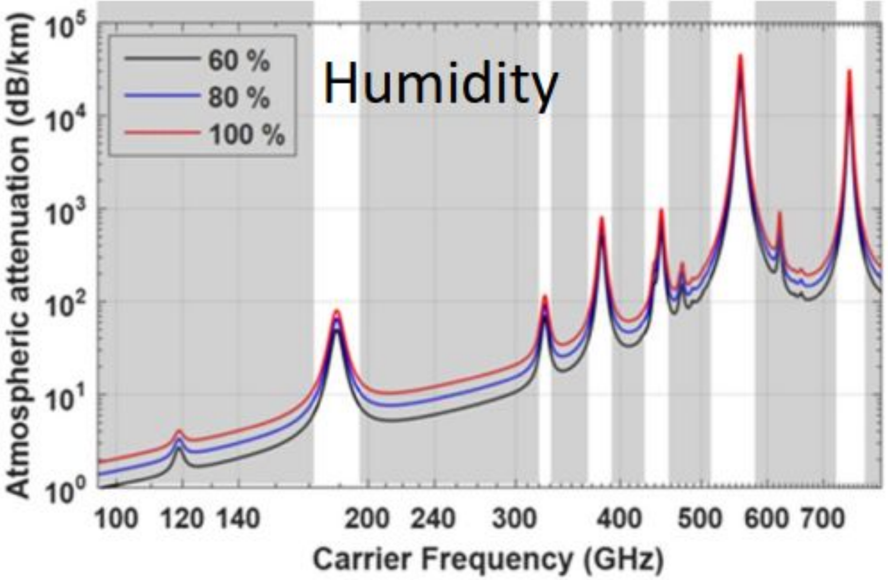
First handheld
phone
(~850 MHz)

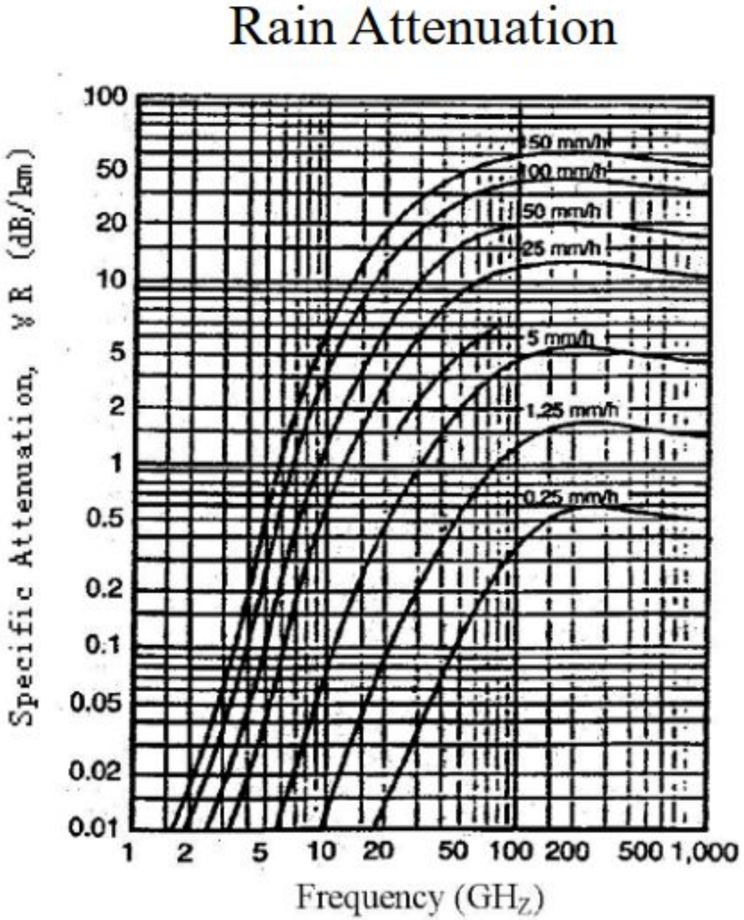
2003

WIFI (~2.4 and
5 GHz)

2019

Foldable Smart
Phone (~3.5 GHz)




**90 Years Just Sub-6 GHz!**


[2] T. S. Rappaport *et al*. "State of the art in 60-GHz integrated circuits and systems for wireless communications," Proceedings of the IEEE, vol. 99, no. 8, pp. 1390–1436, Aug. 2011.

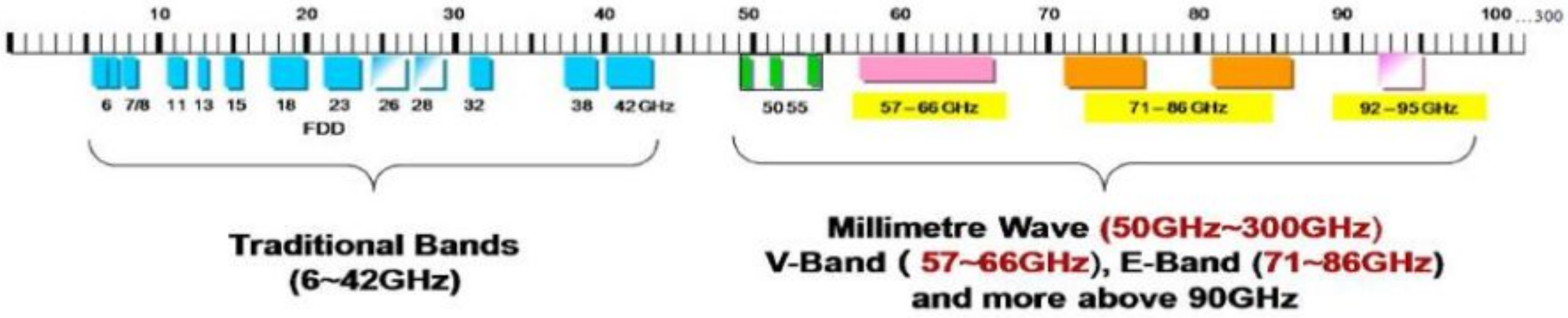
[3] Q. Zhao and J. Li, ''Rain attenuation in millimeter wave ranges,'' in Proc. IEEE Int. Symp. Antennas, Propag. EM Theory, Oct. 2006, pp. 1–4.

[4] mmWave Coalition's NTIA Comments, Filed Jan. 2019. http://mmwavecoalition.org/mmwave-coalition-millimeter-waves/mmwave-coalitions-ntia-comments/

[29] J. Ma et. al., "Channel performance for indoor and outdoor terahertz wireless links," APL Photonics, vol. 3, no. 5, pp. 1–13, Feb. 2018.

# Global Activities on Spectrum Above 95 GHz

- **Europe**: ETSI ISG mWT: studying applications/use cases of millimeter wave spectrum (50 GHz - 300 GHz).


- **ITU-R**: WRC-19 Agenda Item 1.15 will identify applications in the frequency range 275–450 GHz, in accordance with Resolution 767 (WRC-15).


Asia-Pacific Telecommunity (APT) 275–1000 GHz

European Conference of Postal and Telecommunications Administrations (CEPT) 275-1000 GHz

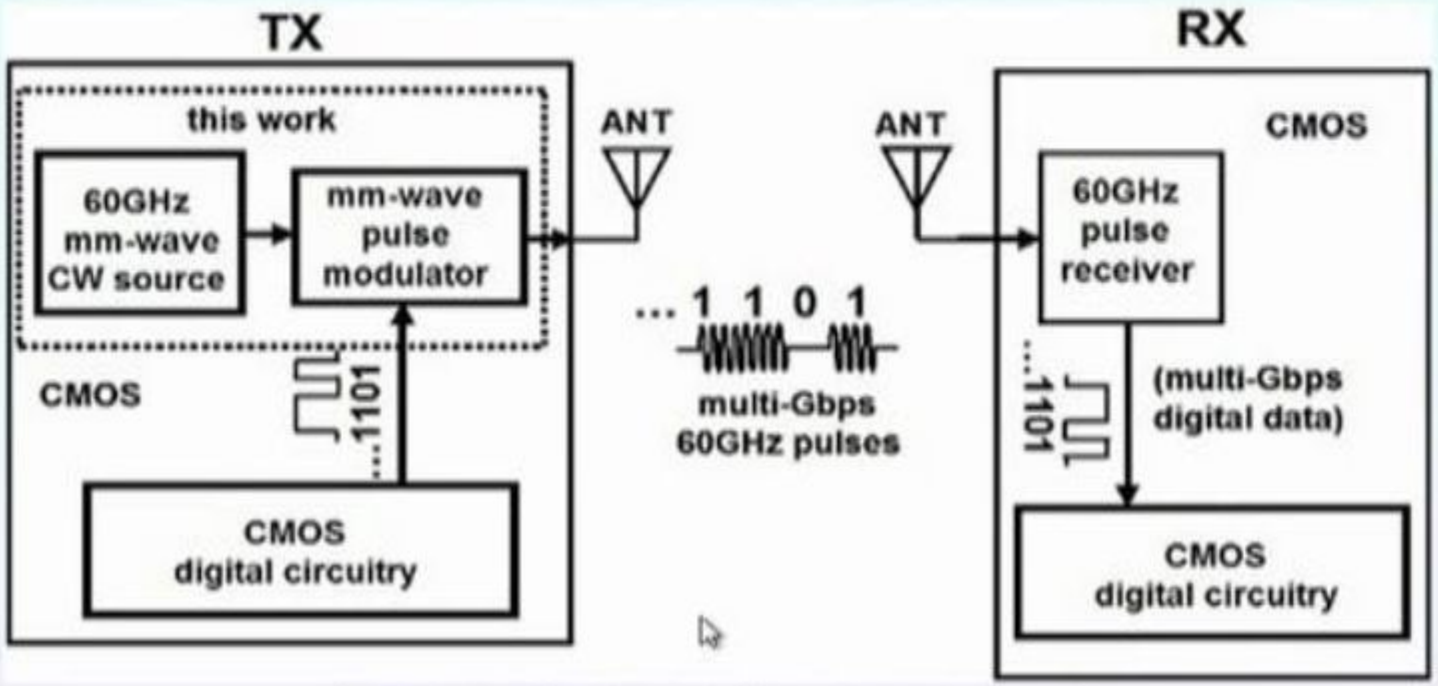
WRC-15

# mmWave & THz Applications

| mmWave & THz Applications—the potential for 6G [1] | |
|---|---|
| Wireless Cognition | Robotic Control [27, 28]<br>Drone Fleet Control [27] |
| Sensing | Air quality detection [5]<br>Personal health monitoring system [6]<br>Gesture detection and touchless smartphones [7]<br>Explosive detection and gas sensing [8] |
| Imaging | See in the dark (mmWave Camera) [9]<br>High-definition video resolution radar [10]<br>Terahertz security body scan [11] |
| Communication | Wireless fiber for backhaul [12]<br>Intra-device radio communication [13]<br>Connectivity in data centers [14]<br>Information shower (100 Gbps) [15] |
| Positioning | Centimeter-level Positioning [9,16] |

[1] T. S. Rappaport, Y. Xing, O. Kanhere, S. Ju, A. Alkhateeb, G. C. Trichopoulos, A. Madanayake, S. Mandal, "Wireless Communications and Applications Above 100 GHz: Opportunities and Challenges for 6G and Beyond (Invited)," IEEE ACCESS, submitted Feb. 2019.

# Millimeter-wave system




A conceptual mobile phone circuit.

Millimeter wave wireless technology presents the potential to offer bandwidth delivery comparable to that of fiber optics, but without the financial and lo challenges of deploying fiber.

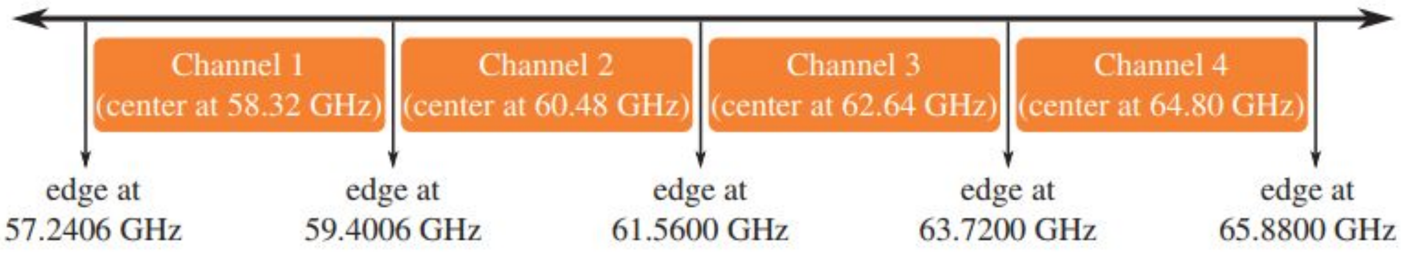



**Figure 9.8** Channelization in IEEE 802.15.3c provides four different channels for mmWave PHY.

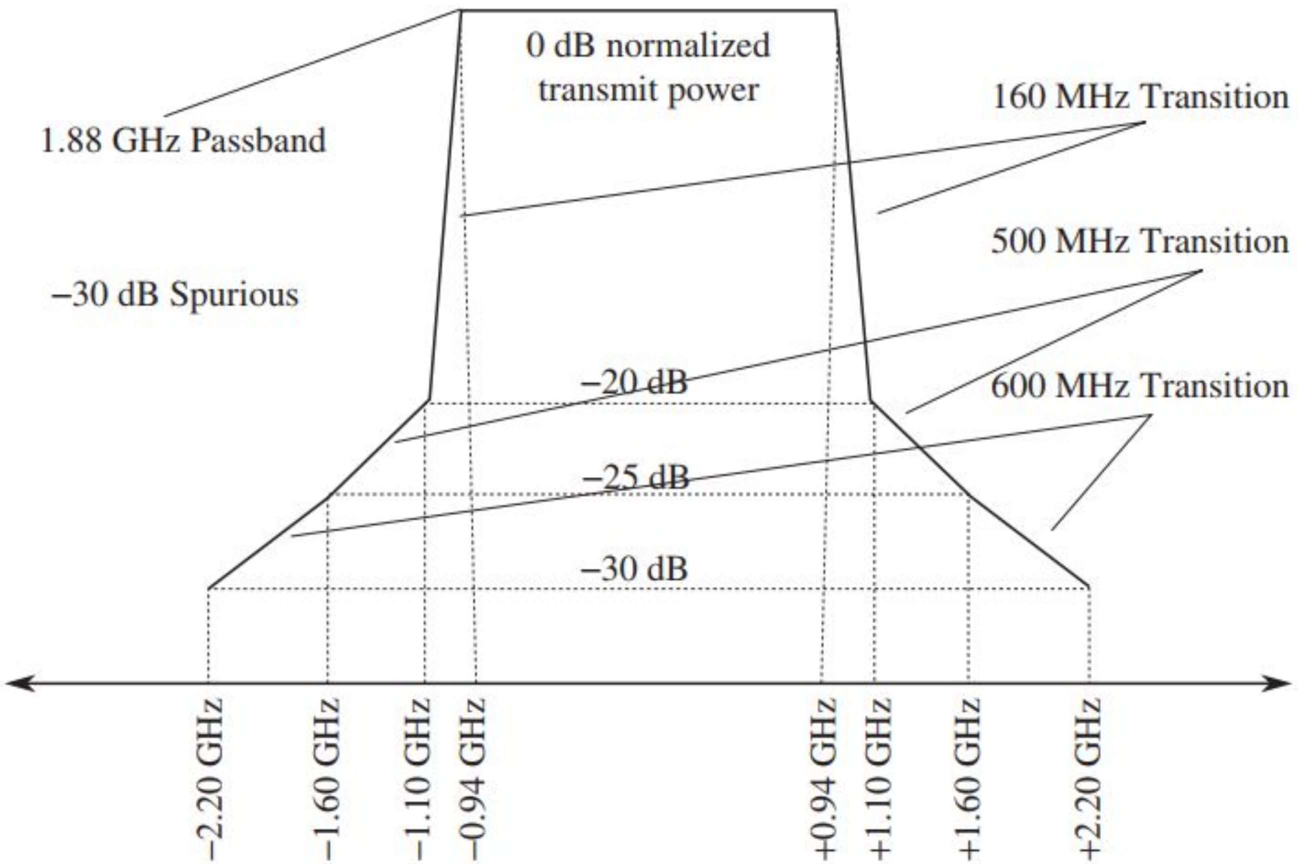
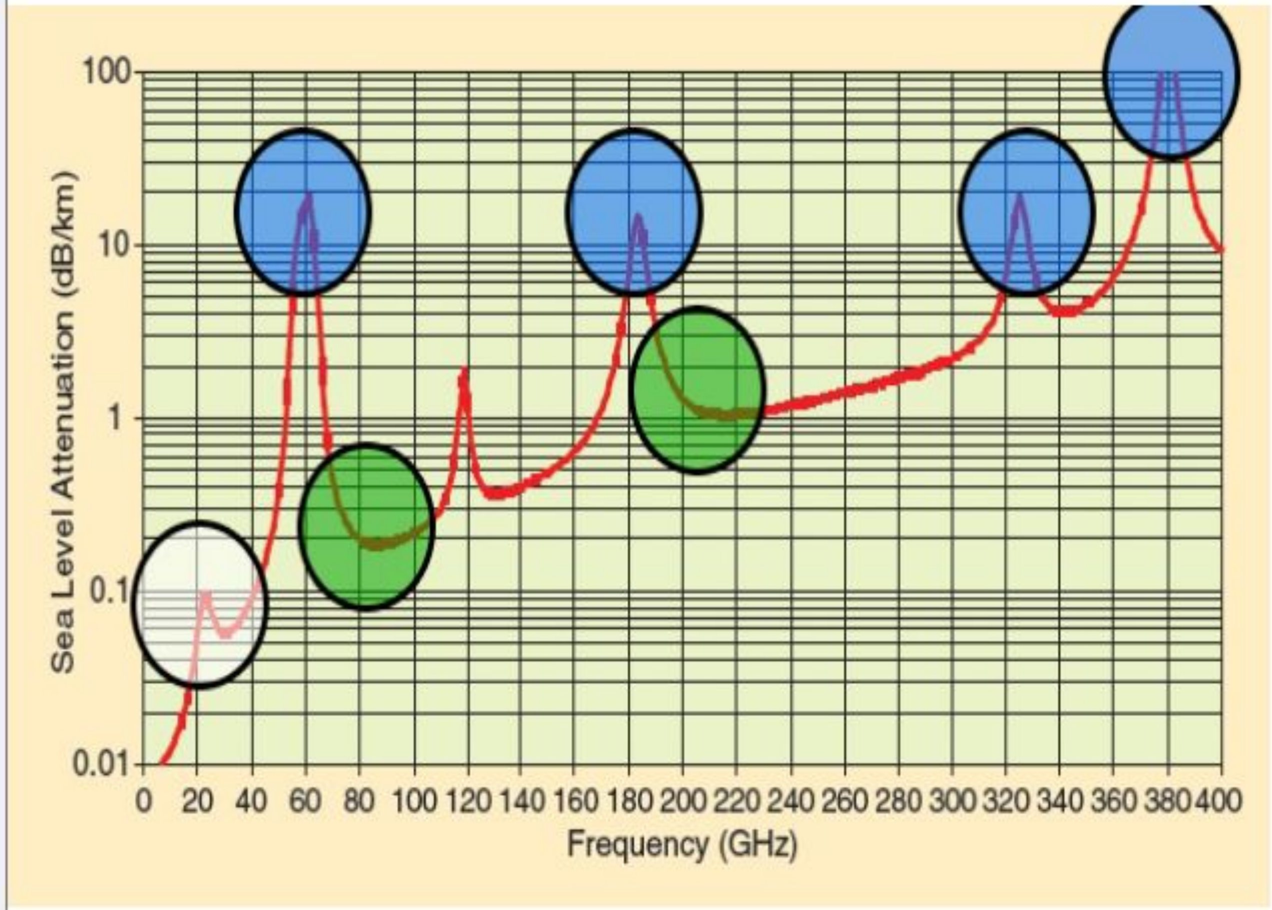



**Figure 9.9** Normalized spectral mask for mmWave PHY transmissions in IEEE 802.15.3c. Transmissions must be attenuated 20 dB ± 0.94 GHz from center frequency, by 25 dB ± 1.10 GHz from center frequency, and by 30 dB ± 2.20 GHz from center frequency. To measure the spectral mask, a resolution bandwidth of 3 MHz and a video bandwidth of 300 kHz is specified in the standard.

# Atmospheric Attenuation: mm-waves

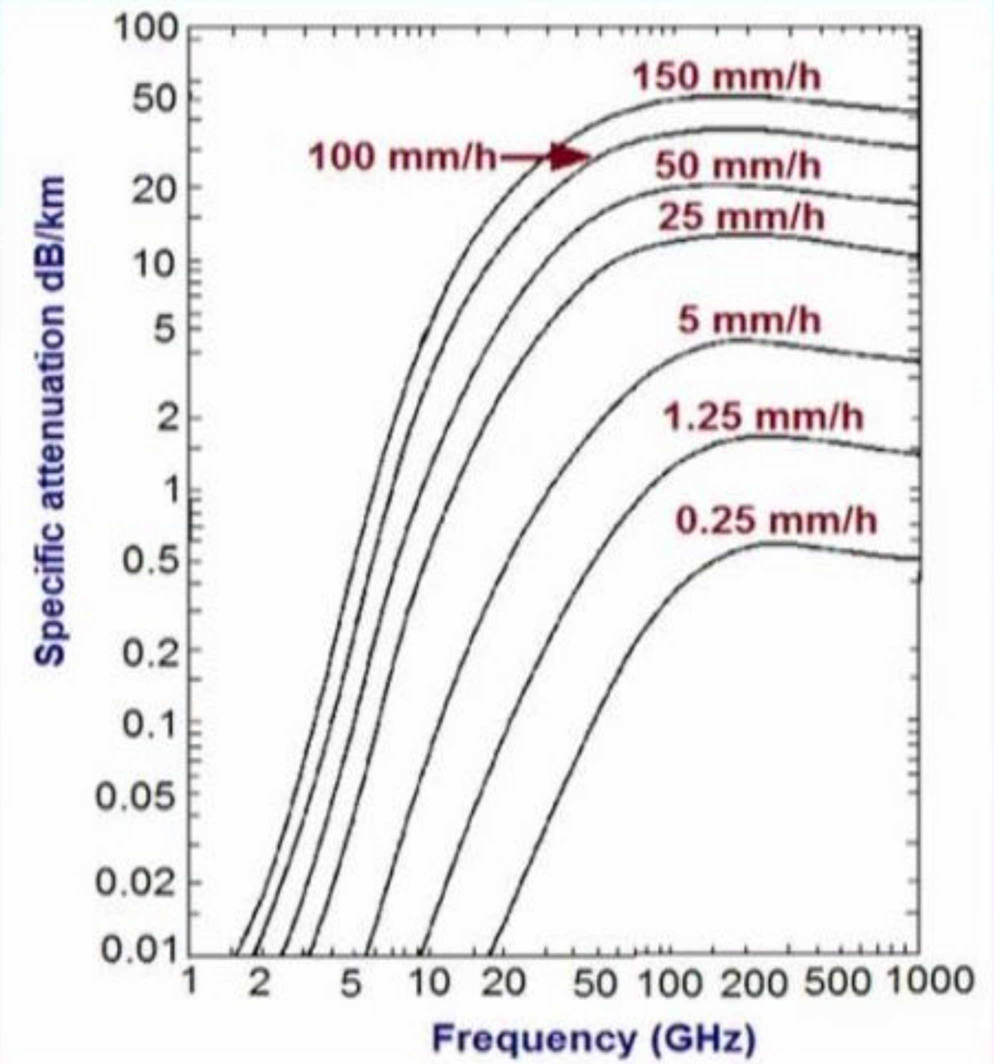



- 0.012 dB over 200 m at 28 GHz
- 0.016 dB over 200m at 38 GHz

- White
  - Current cellular frequencies and low mm-wave
- Blue
  - Short-range indoor communications, whisper radios of the future
    - Higher attenuation
- Green
  - Future backhaul and cellular frequencies
    - Low atmospheric attenuation
    - Multi-GHz Bandwidth
    - Directional Antenna Arrays with Beamsteering
    - CMOS: cost-effective with high frequency limits
    - Atmospherics are challenging

T. S. Rappaport, J. N. Murdock, and F. Gutierrez, "State of the Art in 60-GHz Integrated Circuits and Systems for Wireless Communications," Proceedings of the IEEE, vol. 99, no. 8, pp. 1390–1436, August 2011.

# Rain attenuation




- Attenuation increases with frequency (mm-wave) and rain rate.
- A problem for long distance communication (eg. satellite links).

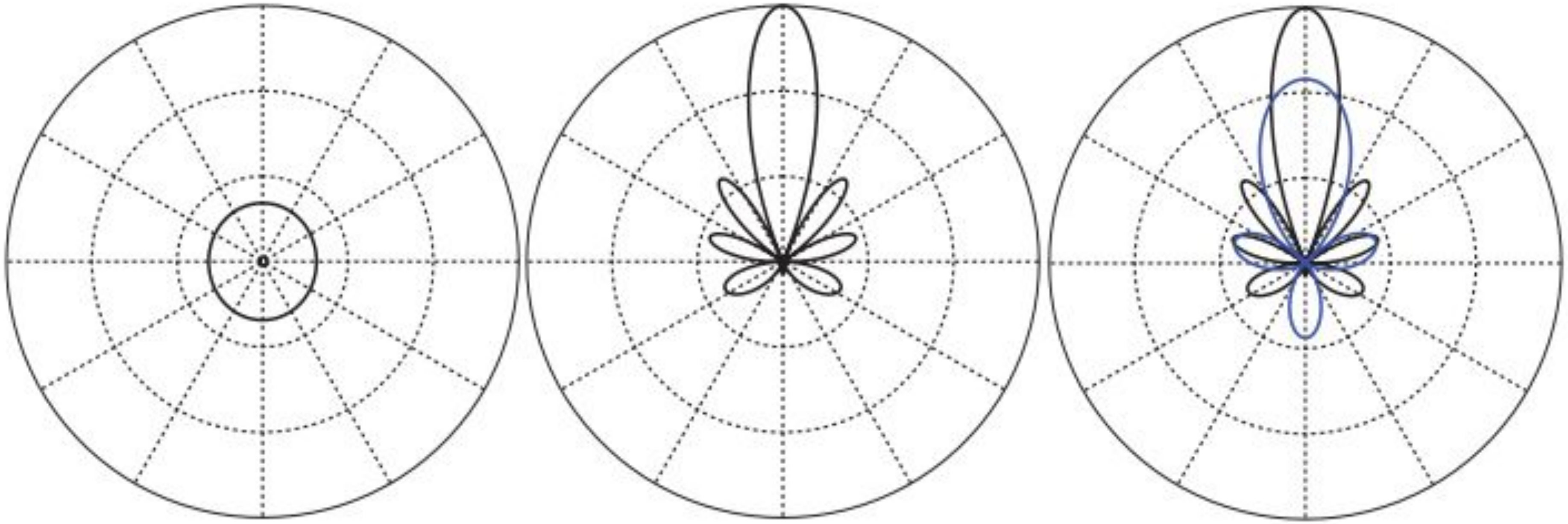
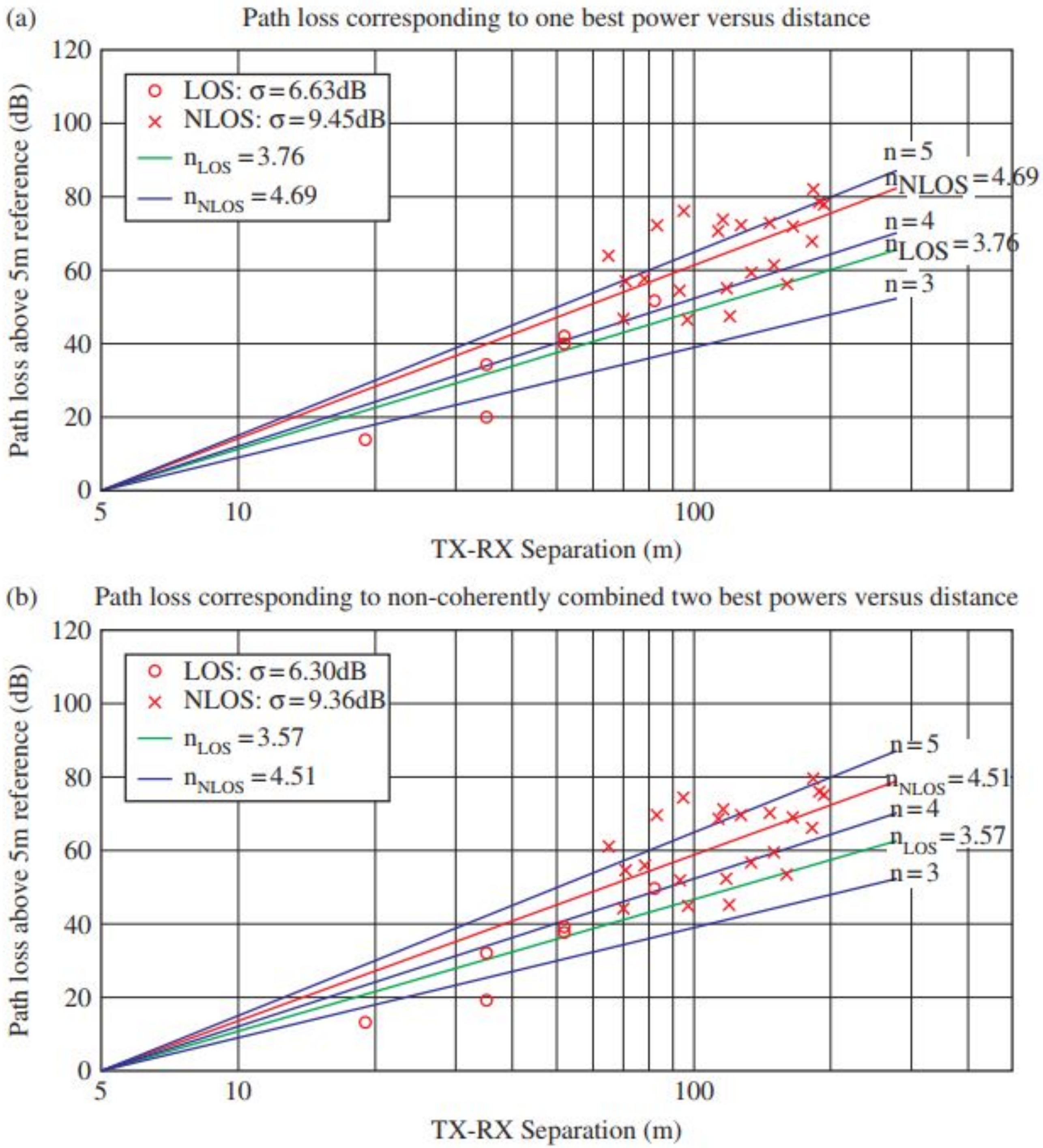
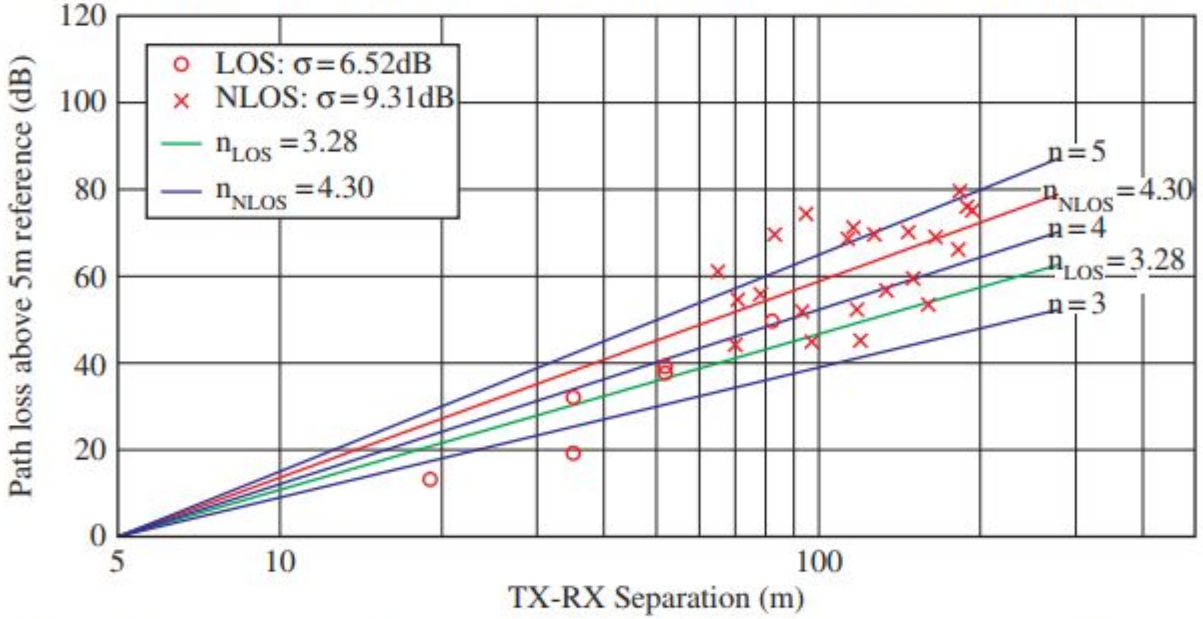


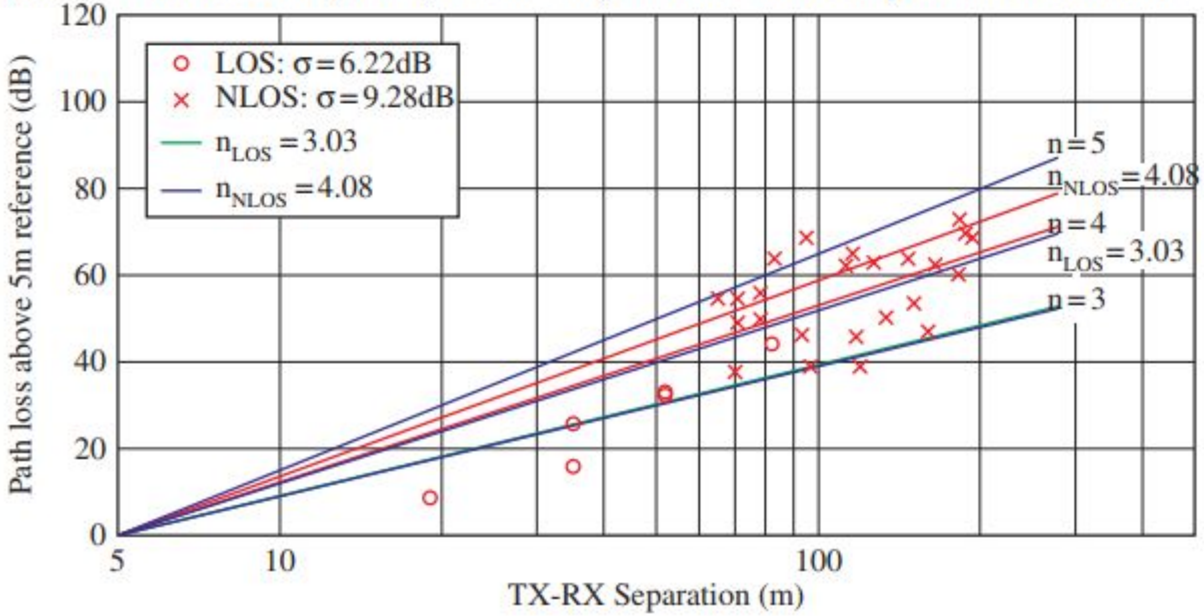
**Figure 8.1** The concept of beamforming. (left) An omnidirectional beam pattern. (middle) A directional beam pattern where a main lobe has high gain presumably in the direction of transmission or reception. (right) Variation of the beam pattern. Wider beams tend to have lower gain than narrower beams but cover a wider area.


(a)
Path loss corresponding to one best power versus distance
LOS: σ=6.63dB
NLOS: σ=9.45dB
$n_{LOS}$ =3.76
$n_{NLOS}$ =4.69
n=5
$n_{NLOS}$=4.69
n=4
$n_{LOS}$=3.76
n=3
Path loss above 5m reference (dB)
TX-RX Separation (m)
(b)
Path loss corresponding to non-coherently combined two best powers versus distance
LOS: σ=6.30dB
NLOS: σ=9.36dB
$n_{LOS}$ =3.57
$n_{NLOS}$ =4.51
n=5
$n_{NLOS}$=4.51
n=4
$n_{LOS}$=3.57
n=3
Path loss above 5m reference (dB)
TX-RX Separation (m)

(c) Path loss corresponding to coherently combined two best powers versus distance




(d) Path loss corresponding to coherently combined three best powers versus distance

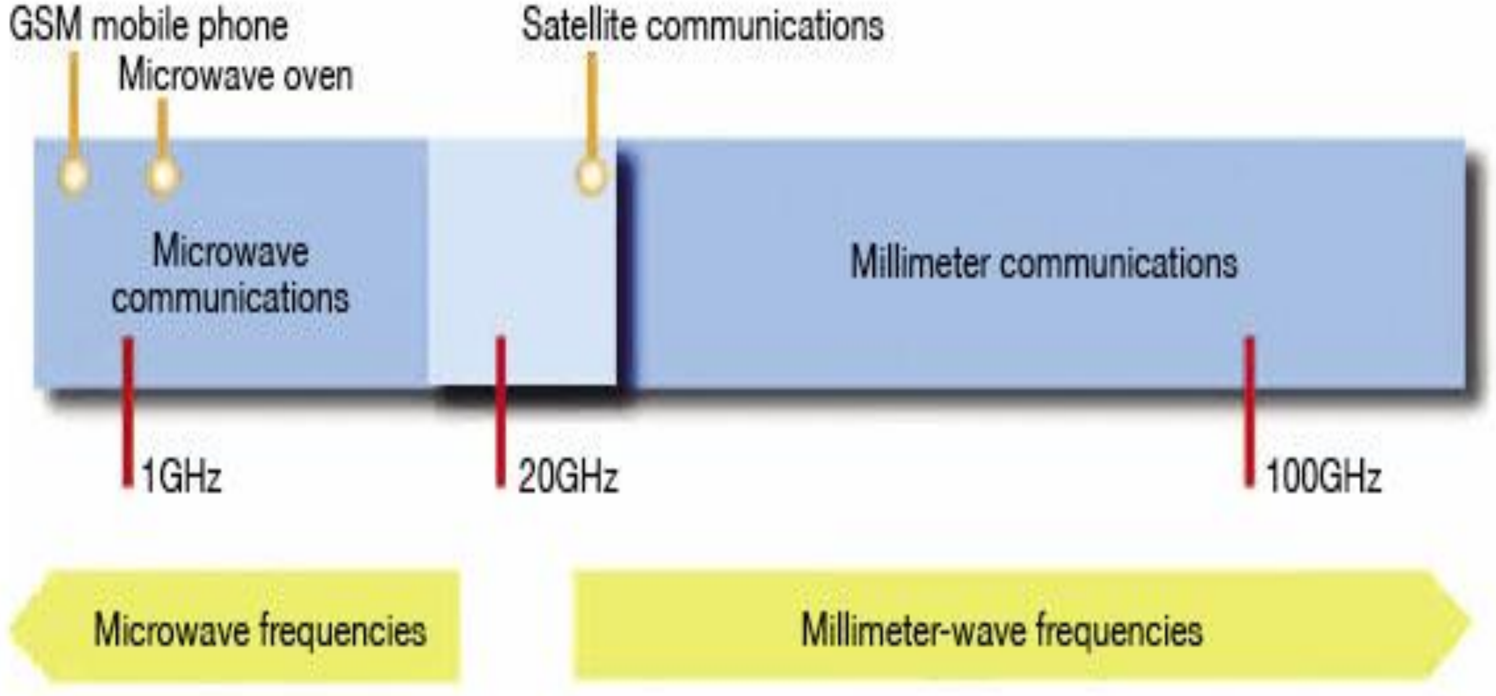


# mm-wave propagation

**Pros and cons:**

- mm-waves have high atmospheric attenuation.
- Rain fade is a serious problem, humidity also has an impact on propagation.
- Surfaces appear rougher so diffused reflection increases.
- Multipath propagation, particularly reflection from indoor walls and surfaces, causes serious fading.
- Doppler shift of frequency can be significant even at pedestrian speeds→ in portable devices shadowing due to the human body is a problem.
- mm-waves travel solely by line-of-sight, and are blocked by building walls and attenuated by foliage (a few km)→ densely packed communications networks (WPAN) through frequency reuse.
- Show "optical" propagation characteristics → can be reflected and focused by small metal surfaces, diffracted by building edges.
- The small wavelength allows modest size antennas to have a small beam width, increasing frequency reuse potential, small circuit size.
- Potential applications: very high resolution radar, communication links> 10 GBPS.

# Millimeter wave frequency spectrum

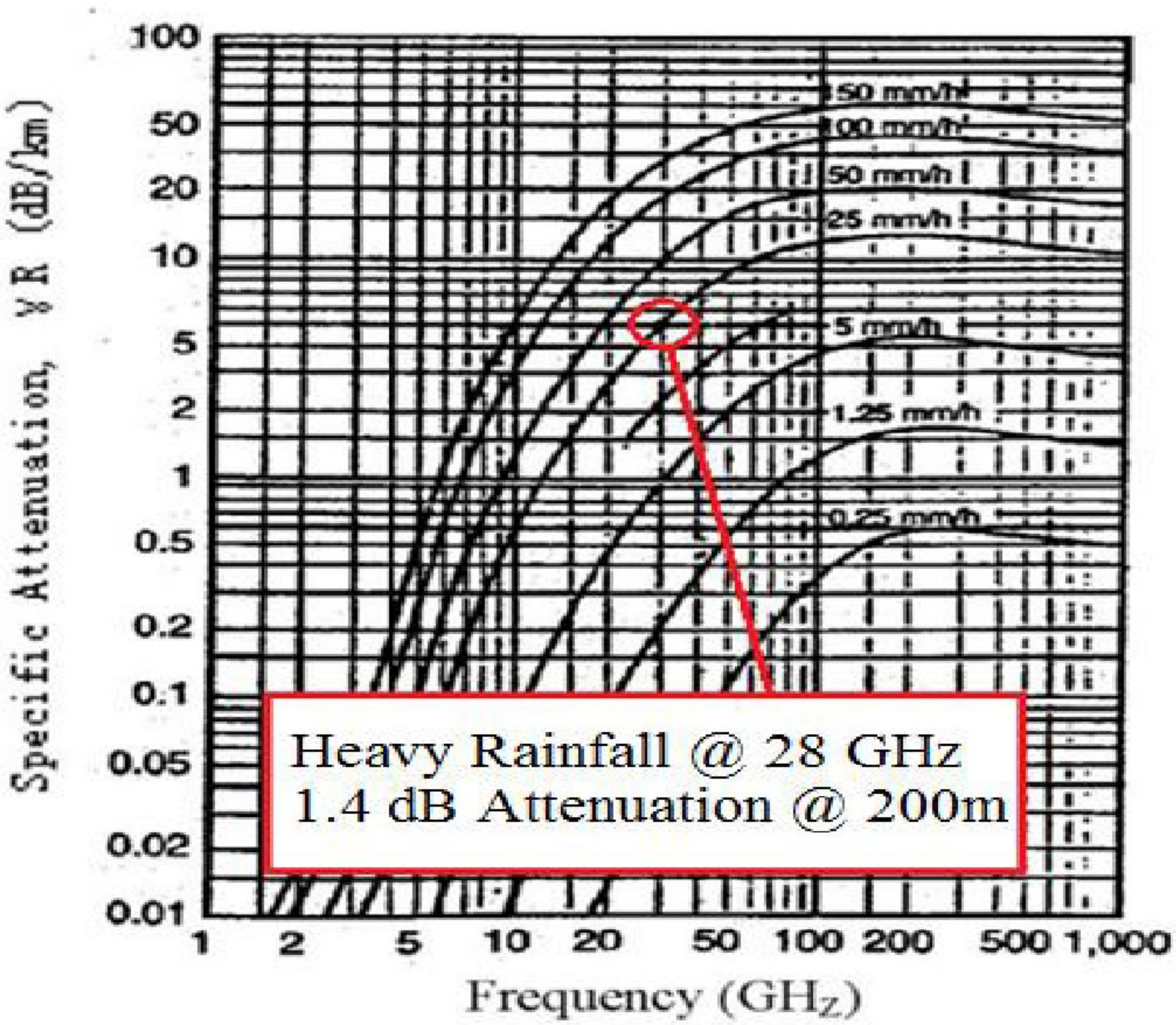


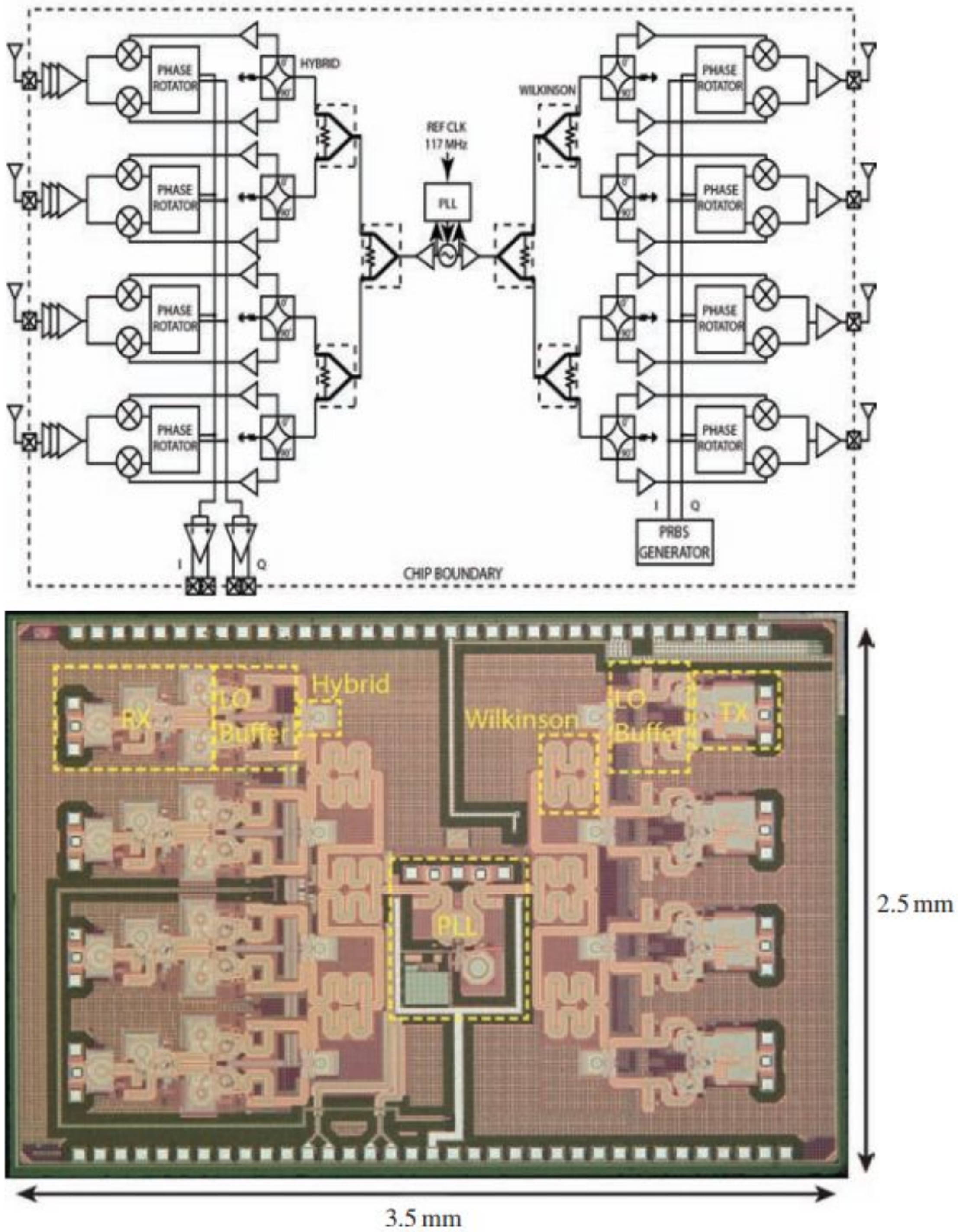

Communication in Millimeter-Wave Frequencies

- Mm-wave spectrum would allow service providers to significantly expand the channel bandwidths far beyond the present 20 MHz channels used by 4G customers.
- By increasing the RF channel bandwidth for mobile radio channels, the data capacity is greatly increased, while the latency for digital traffic is greatly decreased, thus supporting much better internet based access and applications that require minimal latency.
- Mm-wave frequencies, due to the much smaller wavelength, may exploit polarization and new spatial processing techniques, such as **massive MIMO and adaptive beam forming**.
- the mm-wave spectrum will have spectral allocations that are relatively much closer together, making the propagation characteristics of different mm-wave bands much more comparable and ``homogenous''.

- A common myth in the wireless engineering community is that rain and atmosphere make mm-wave spectrum useless for mobile communications.
- However, when one considers the fact that today's cell sizes in urban environments are on the order of 200 m, it becomes clear that mm-wave cellular can overcome these issues.
- Figure shows the rain attenuation and atmospheric absorption characteristics of mm-wave propagation.
- It can be seen that for cell sizes on the order of 200 m, atmospheric absorption does not create significant additional path loss for mm-waves, particularly at 28 GHz and 38 GHz.
- Only 7 dB/km of attenuation is expected due to heavy rainfall rates of 1 inch/hr for cellular propagation at 28 GHz, which translates to only 1.4 dB of attenuation over 200 m distance.

**Rain attenuation in db/km across frequency at various rain fall rates**






Block diagram (top) and die photo (bottom) of an integrated circuit with four transmit and receive channels, including the voltage-controlled oscillator, phase-locked loop, and local oscillator distribution network

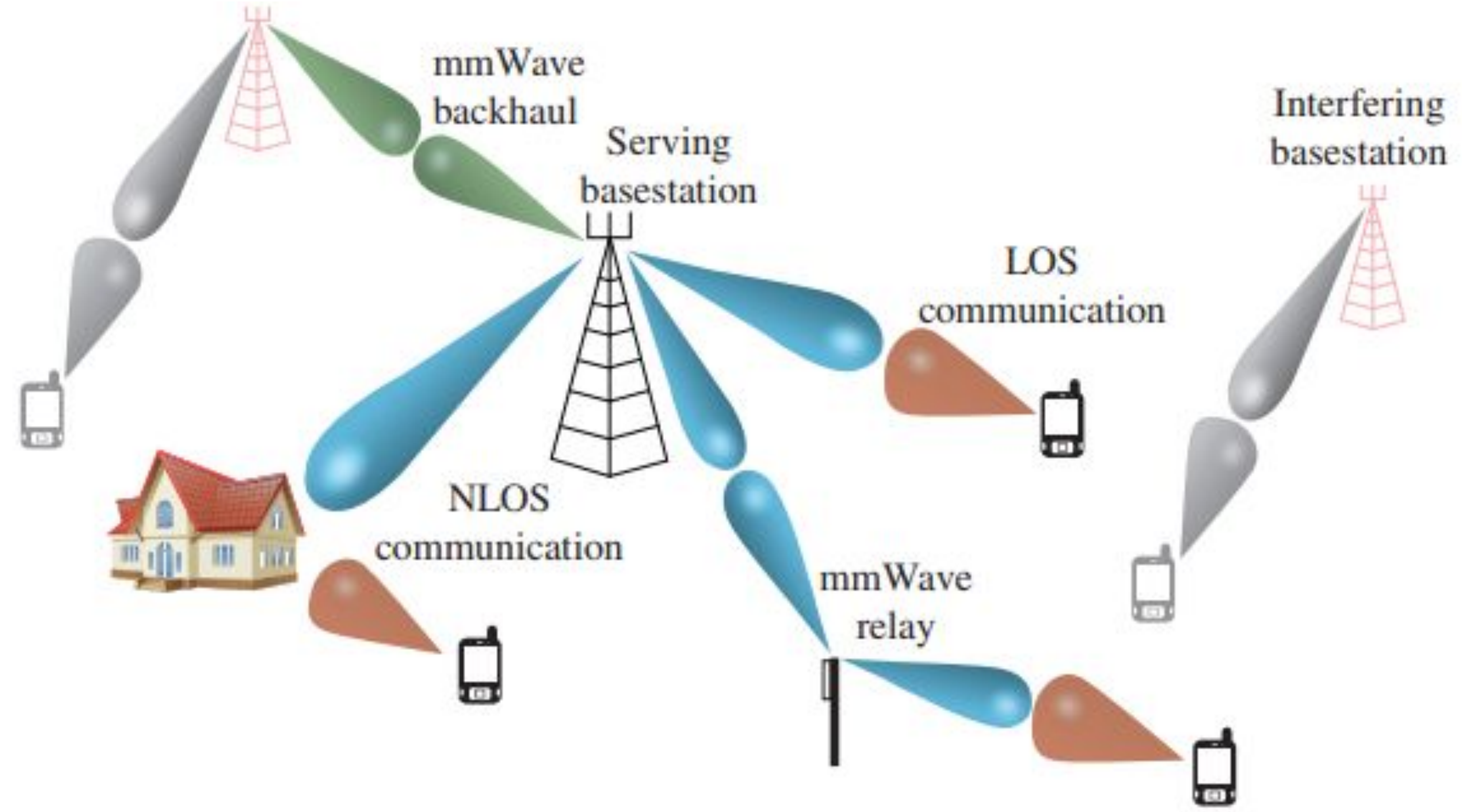
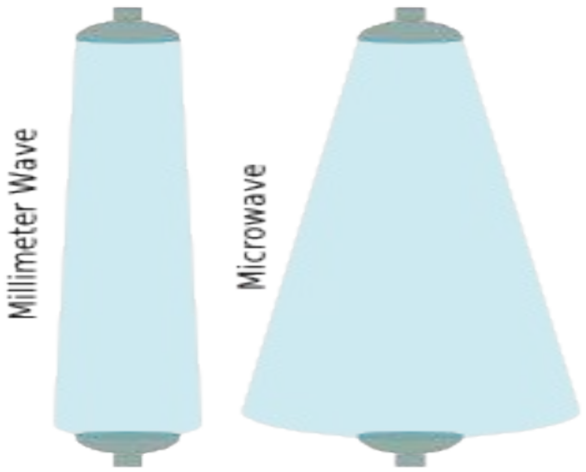



**Figure 1.12** Illustration of a mmWave cellular network. Base stations communicate to users (and interfere with other cell users) via LOS, and NLOS communication, either directly or via heterogeneous infrastructure such as mmWave UWB relays.

# Parameter Affected By mm-wave

- **BANDWIDTH:-**The main benefit that millimeter Wave technology has over RF frequencies is the spectral bandwidth of 5GHz being available in these ranges, resulting in current speeds of 1.25Gbps Full Duplex with potential throughput speeds of up to 10Gbps Full Duplex being made possible.
- **SECURITY**:-Since millimeter waves have a narrow beam width and are blocked by many solid structures they also create an inherent level of security. In order to sniff millimeter wave radiation a receiver would have to be setup very near, or in the path of, the radio connection. The loss of data integrity caused by a sniffing antenna provides a detection mechanism for networks under attack. Additional measures, such as cryptographic algorithms can be used that allow a network to be fully protected against attack.

- **BEAM WIDTH INTERFERENCE RESISTANCE**:-Millimeter wave signals transmit in very narrow focused beams which allows for multiple employments in close range using the same frequency ranges. This allows Millimeter wave ideal for Point-to-Point Mesh, Ring and dense Hub & Spoke network topologies where lower frequency signals would not be able to cope before cross signal interference would become a significant limiting factor.



# Advantages And Limitation Of mm-wave

ADVANTAGES:-

- Millimeter wave's larger bandwidth is able to provide **higher transmission rate, capability of spread spectrum and is more immune to interference.**
- Extremely high frequencies allow **multiple short-distance** (I.e. multiple TX can be placed in nearby location to each other) usages at the same frequency without interfering each other but It requires the narrow beam width. **For the same size of antenna, when the frequency is increased, the beam width is decreased.**
- It reduces hardware size, i.e. higher the frequency is, the smaller the antenna size can be used.

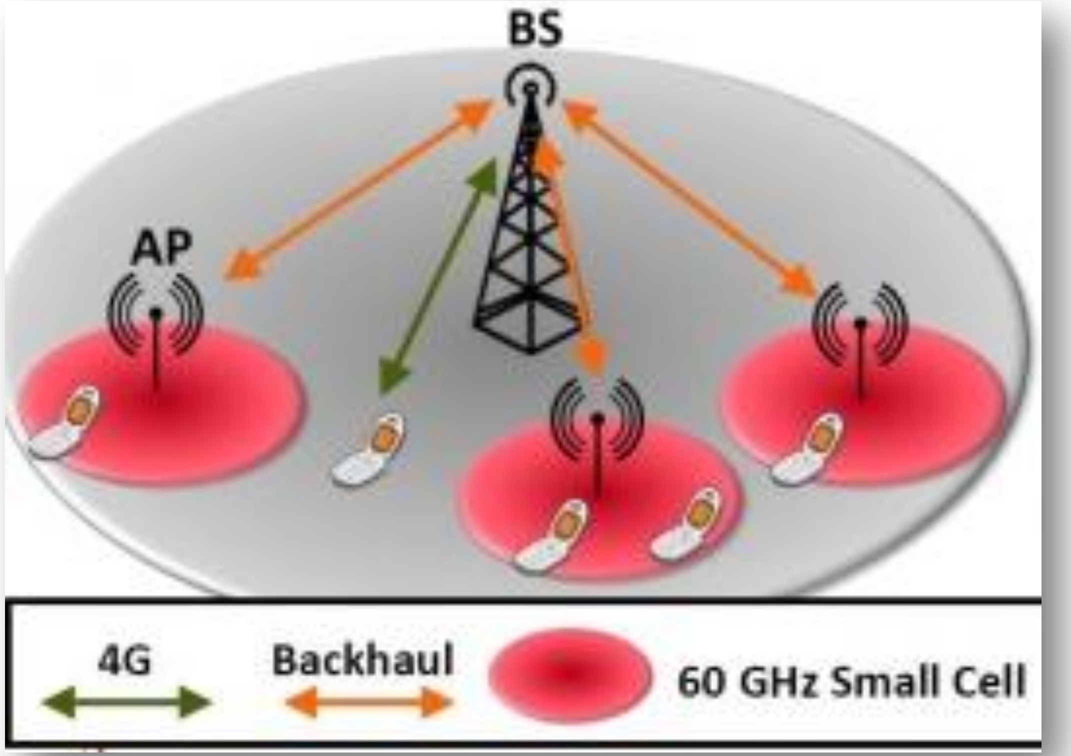
# LIMITATIONS

- Higher costs in manufacturing of **greater precision hardware due to components with smaller size.**
- At extremely high frequencies, **there is significant attenuation**. Hence millimeter waves can hardly be used for long distance applications.
- The penetration power of mm-wave through objects such concrete walls is known less.
- There are interferences with oxygen & rain at higher frequencies therefore further research is going on to reduce this.

# Applications of Mm wave communication

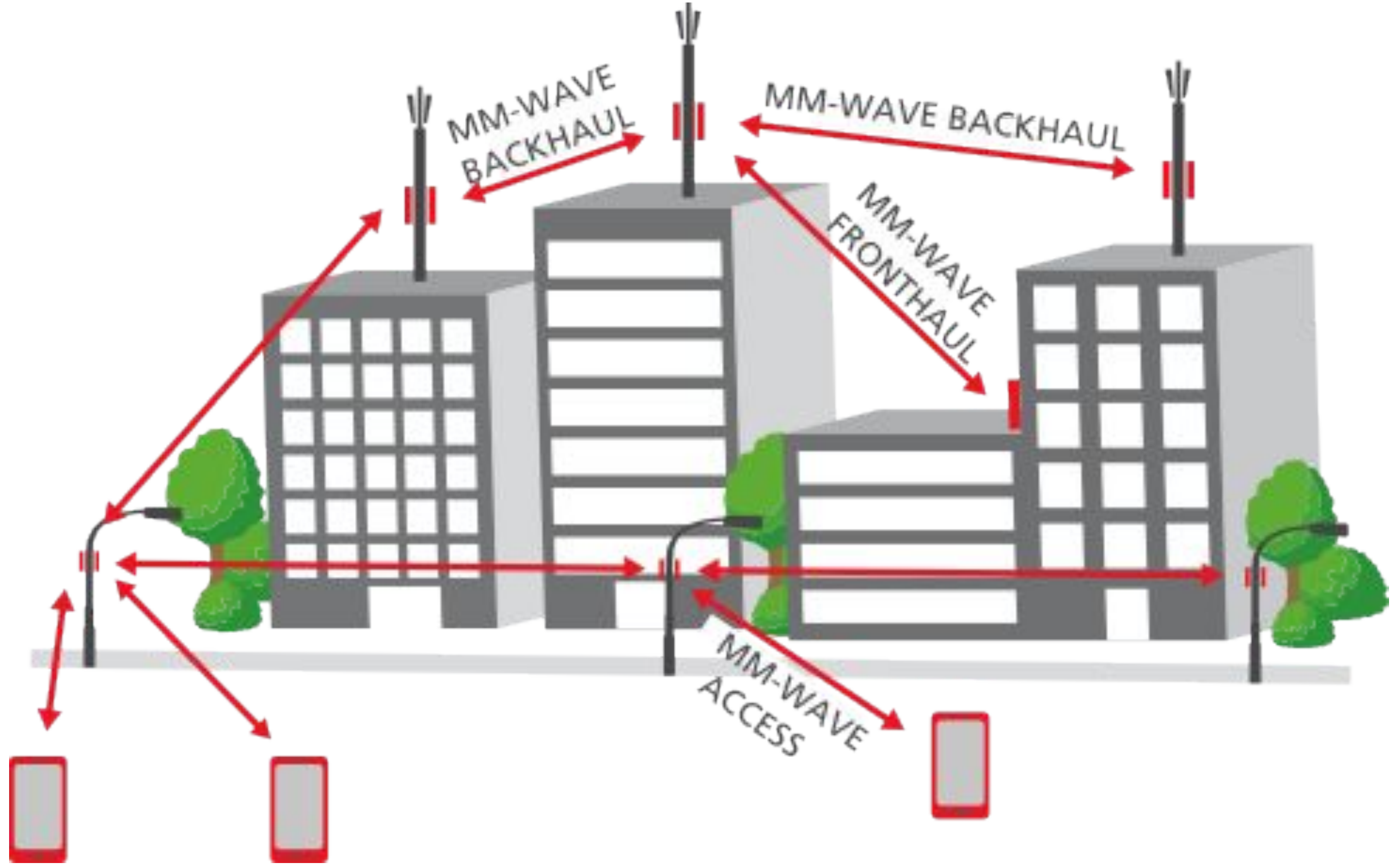
## I. Small Cell Access :

Small cells deployed underplaying the macro cells and provide solution for the capacity enhancement in the 5G networks. With huge bandwidth, mm Wave small cells are able to provide the gigabit rates. Small cells encrypt all voice and data sent and received.



- **II. Wireless Backhaul**

With small cells densely deployed in the next generation of cellular systems (5G), it is costly to connect base stations (BSs) to the other BSs and to the network by fiber based backhaul .In contrast, high speed wireless backhaul with low cost, flexible, and easier to deploy. With huge bandwidth available, wireless backhaul in mm Wave bands, such as the 60 GHz band and E-band (71–76 GHz and 81–86 GHz), provides several-Gbps data rates and can be a promising backhaul solution for small cells. The Eband backhaul provides the high speed transmission between the small cell base stations (BSs) or between BSs and the gateway.


MM-WAVE BACKHAUL
MM-WAVE BACKHAUL
MM-WAVE FRONTHAUL
MM-WAVE ACCESS

### III. millimeter wave propagation

The propagation characteristics of millimeter wave bands are very different to those below 4GHz. Typically distances that can be achieved are very much less and the signals do not pass through walls and other objects in buildings.

Typically millimeter wave communication is likely to be used for outdoor coverage ranges between 200 - 300 meters.

Often these  millimeter wave small cells may use beamforming techniques to target the required user equipment and also reduce the possibility of reflections.

# References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